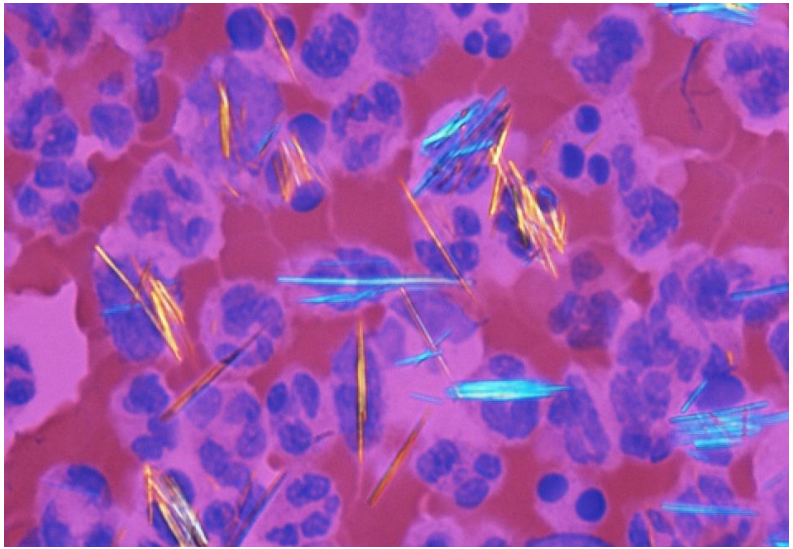
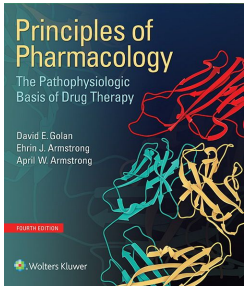
PHARMACOLOGY: Gout Treatment

Kristen Mitchell, PhD
Office 348A
kmitchell@icom.edu

RENL Fall 2026



Resources



Golan, DA, Armstrong EJ, Armstrong AW. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, 5e, Philadelphia, PA: Wolters Kluwer.

Chapter 56: Pharmacology of Gout

***** You may upload my lecture materials into Gemini/Notebook.***

Session Objectives

1. Describe four ways CKD can alter the safety and efficacy of gout drugs.
2. Explain why NSAIDs, colchicine, allopurinol, anakinra, and probenecid require caution or avoidance in CKD.
3. Identify gout drugs that are largely unaffected by reduced kidney function.
4. Select appropriate flare treatment and urate-lowering therapy for a patient with CKD.

How this lecture differs from previous lectures on gout:

	MSKS Pharm	RENL Biochem (Dr. Fologea)	Today: RENL Pharm
Focus	Gout drugs	Uric acid production and excretion	Gout drugs in patients with reduced kidney function
Key content	Flare vs. urate-lowering drugs; MOA; adverse effects	Purine degradation; xanthine oxidase; uric acid solubility; underexcretion vs. overproduction	Which drugs become unsafe, which become ineffective, which interactions matter
Assumed kidney function	Normal	Not the focus	Reduced, acutely injured, dialysis, or transplant
Key question	What does the drug do?	Why is urate high?	Is this drug still appropriate for this patient?

- ***Gout and reduced kidney function often occur together.***
- ***Kidney dysfunction changes how gout drugs are cleared, how well they work, and how safe they are.***

Chronic Kidney Disease Is Staged by eGFR

- **Chronic kidney disease (CKD):** caused by conditions that damage kidneys over time, reducing their ability to filter waste and extra fluid from blood. Leading causes:
  - Diabetes (type I and II)
  - Hypertension
- **eGFR** (estimated glomerular filtration rate) is calculated from serum creatinine and reported in mL/min/1.73 m^2
- **Higher stage = lower kidney function**
- Gout drug recommendations are written in terms of CKD stage.

Kidney disease stage and eGFR

Stage	Description	eGFR
1	Possible kidney damage (e.g., protein in the urine) with **normal** kidney function	90 or above
2	Kidney damage with **mild loss** of kidney function	60 to 89
3a	**Mild to moderate** loss of kidney function	45 to 59
3b	**Moderate to severe** loss of kidney function	30 to 44
4	**Severe loss** of kidney function	15 to 29
5	Kidney **failure**	Less than 15

Review: Drugs Used to Treat Gout

Goal	Class	Drugs	Mechanism
Treat flares and prevent flares when starting urate lowering	**NSAIDs**	Naproxen, indomethacin, ibuprofen	Inhibit COX-1/2; ↓ prostaglandins
	Colchicine	Colchicine	Binds tubulin; ↓ neutrophil migration and ↓ NLRP3 inflammasome activation
	Glucocorticoids	Prednisone, prednisolone, triamcinolone (intra-articular)	↓ inflammatory gene expression
	IL-1 inhibitors	Anakinra (off-label), canakinumab	Block IL-1 signaling (anakinra: IL-1 receptor; canakinumab: IL-1β)
Lower urate (recurrent gout, tophi)	**Xanthine oxidase inhibitors**	Allopurinol, febuxostat	Inhibit xanthine oxidase; ↓ uric acid production
	Uricosuric	Probenecid	Inhibits URAT1; ↑ urate excretion
	Uricase	Pegloticase	Converts uric acid to allantoin

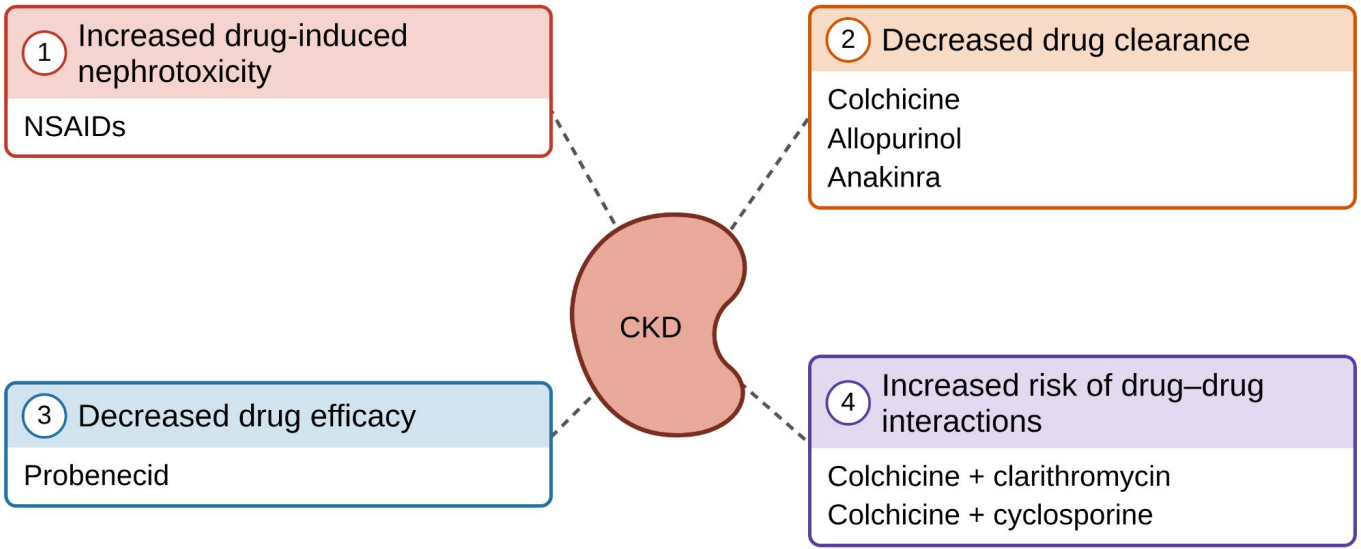
Mr. R is a 61-year-old man with CKD stage 4. Over the past year, his serum creatinine has been stable at about 2.5 mg/dL (0.6–1.2 mg/dL), and his serum urate has ranged from 9 to 10 mg/dL (3.0–8.2 mg/dL). He has had two gout flares in the past year, each treated with a short course of oral prednisone. Three days ago he developed a third flare and has been taking over-the-counter ibuprofen for the pain. On examination, his right first metatarsophalangeal joint is red, warm, and swollen, and a small tophus is present on his left elbow. Today his serum creatinine is 3.3 mg/dL, indicating worsening kidney function.

Why did his creatinine rise?

How would you treat this flare?

Should he start urate-lowering therapy?

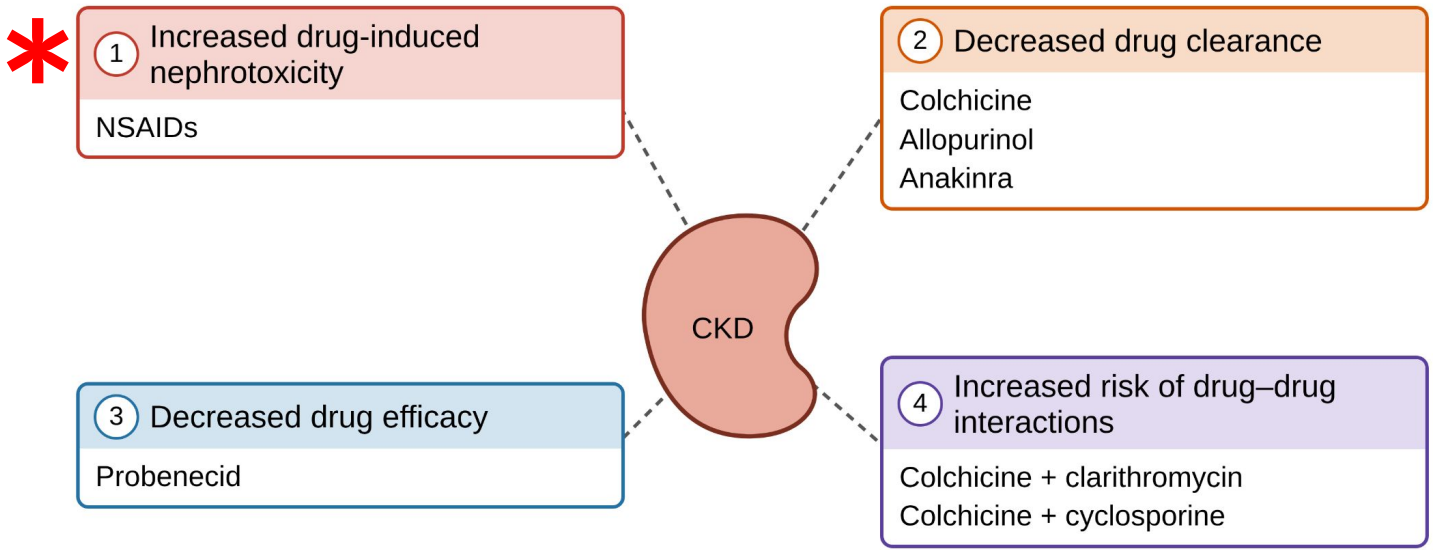
How CKD Can Affect the Safety and Efficacy of Gout Drugs



Drugs that are largely unaffected by reduced kidney function

Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

How CKD Can Affect the Safety and Efficacy of Gout Drugs

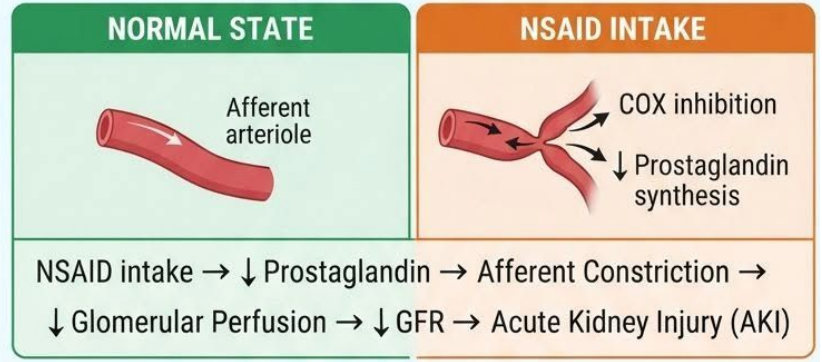


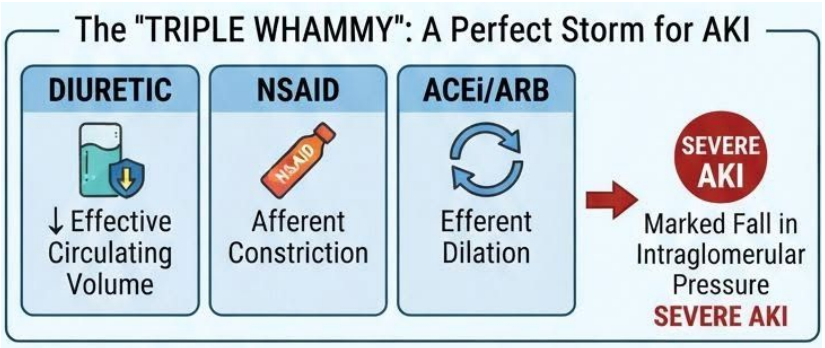
Drugs that are largely unaffected by reduced kidney function

Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

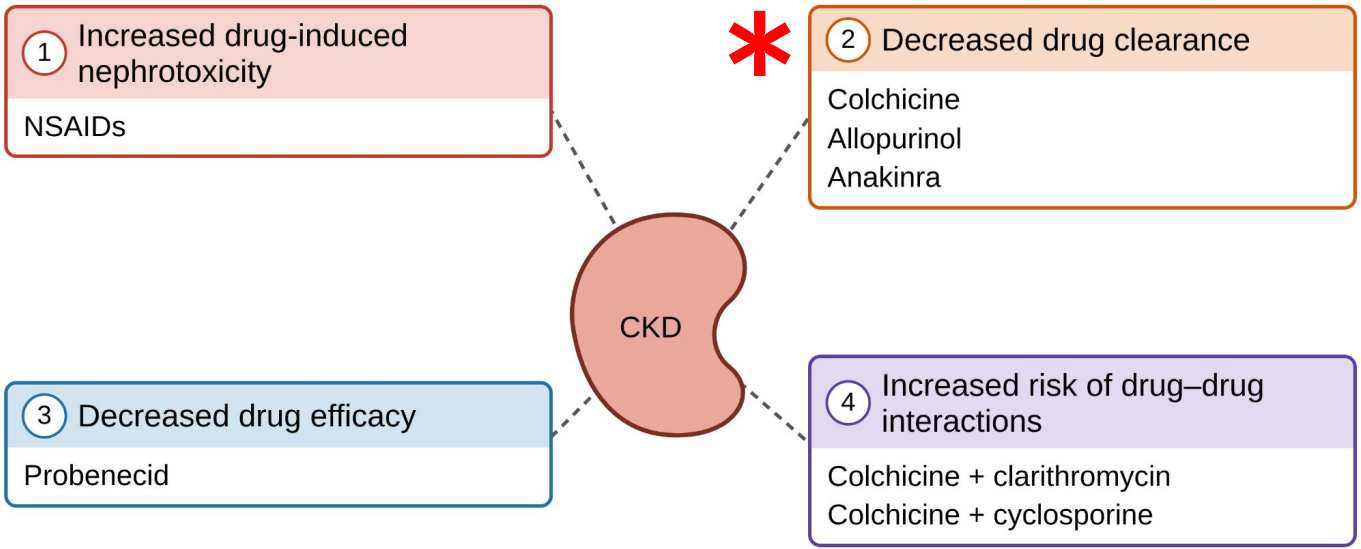
NSAIDs Can Cause Acute Kidney Injury

- In CKD, renal prostaglandins dilate the **afferent** arteriole to maintain blood flow into the glomerulus.
- NSAIDs block prostaglandin synthesis, which lowers GFR and can cause **acute kidney injury (AKI),** a sudden decline in kidney function.
- Other renal adverse effects include sodium and water retention, hyperkalemia, and **interstitial nephritis** (an immune reaction).
- Includes COX-2 selective NSAIDs (celecoxib).
- **Avoid NSAIDs in CKD stage 3 or worse.**





How CKD Can Affect the Safety and Efficacy of Gout Drugs

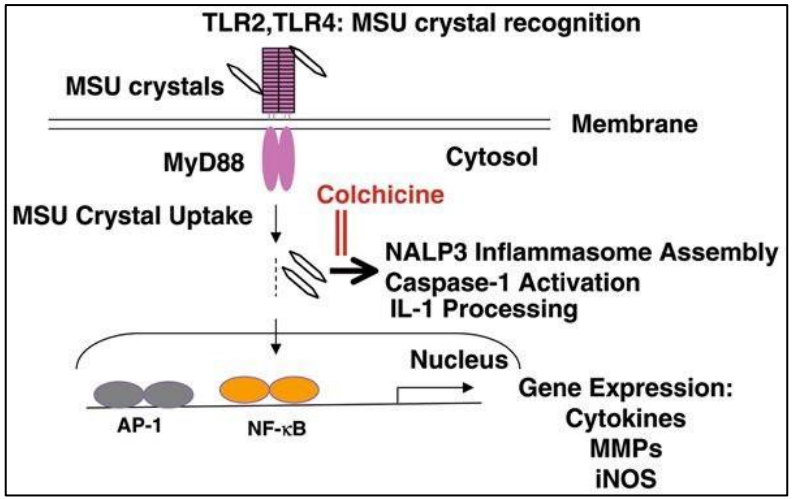


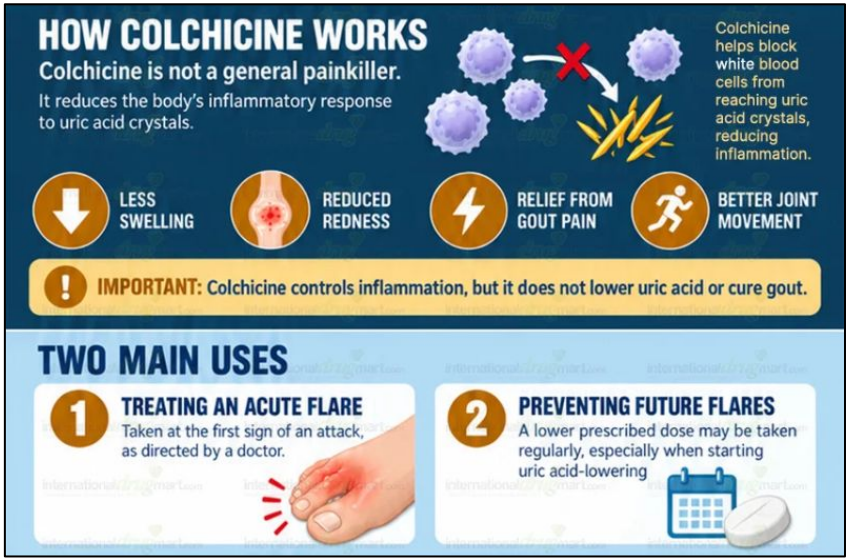
Drugs that are largely unaffected by reduced kidney function

Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

Decreased Clearance: Colchicine

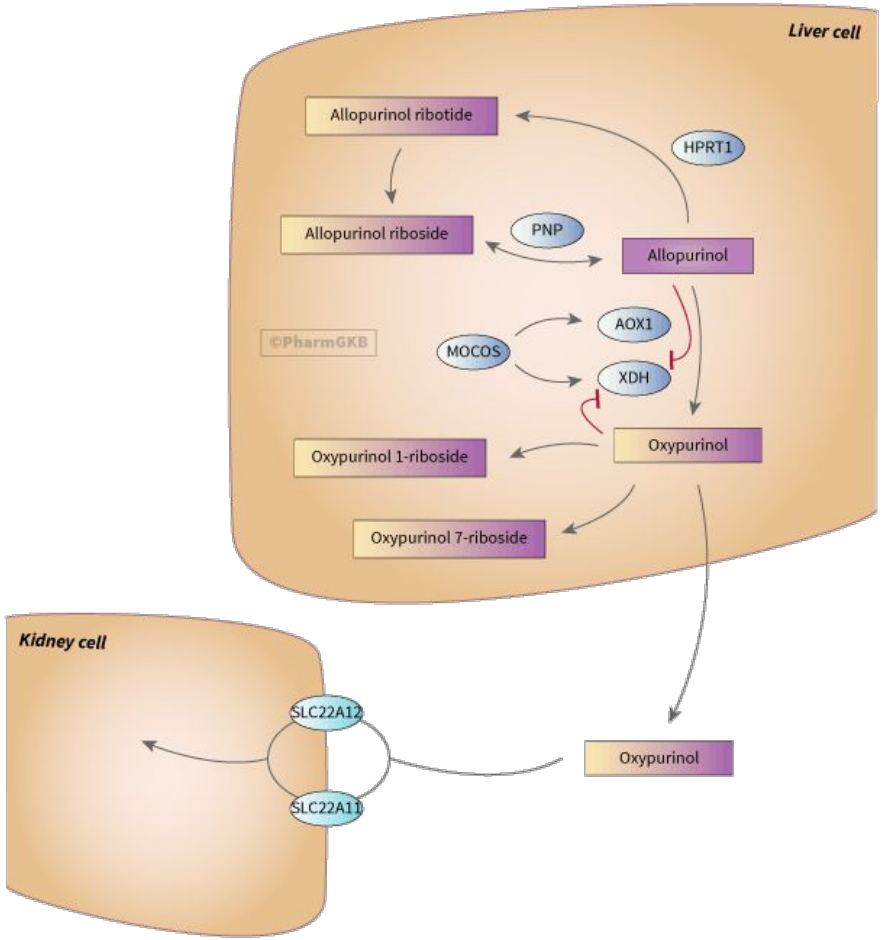
- MOA (review): binds tubulin, which reduces neutrophil migration and inflammasome activation.
- **Narrow therapeutic index**, partly cleared by the kidney.
- In CKD, colchicine accumulates and toxicity rises.
- **Not removed by dialysis.**
- Toxicity: GI symptoms first, then **myoneuropathy** (proximal weakness, elevated CK), **rhabdomyolysis**, and **bone marrow suppression**
- Use with caution in CKD; avoid repeated courses in CKD stage 4–5.





Decreased Clearance: Allopurinol

- **First-line urate-lowering drug, including in CKD.**
- Allopurinol is converted to its active metabolite, **oxypurinol**, which is cleared by the kidney.
- In CKD, reduced GFR decreases oxypurinol clearance, so it **accumulates.**
- Higher oxypurinol levels increase the risk of **hypersensitivity reactions**, including interstitial nephritis
- In CKD: **start low, increase gradually,** and **monitor** serum urate, kidney function, and for rash.



Allopurinol Hypersensitivity Syndrome

- Rare but potentially fatal; usually begins within the first few months of therapy.
- Features: rash, fever, and injury to organs, such as the kidney and liver.
- Risk factors: **CKD**, HLA-B*58:01, high starting dose.
- Test for **HLA-B*58:01** before starting in higher-risk populations: African American descent and Asian descent (Han Chinese, Korean or Thai).
- Rash with fever or other symptoms: stop allopurinol immediately and never restart it.



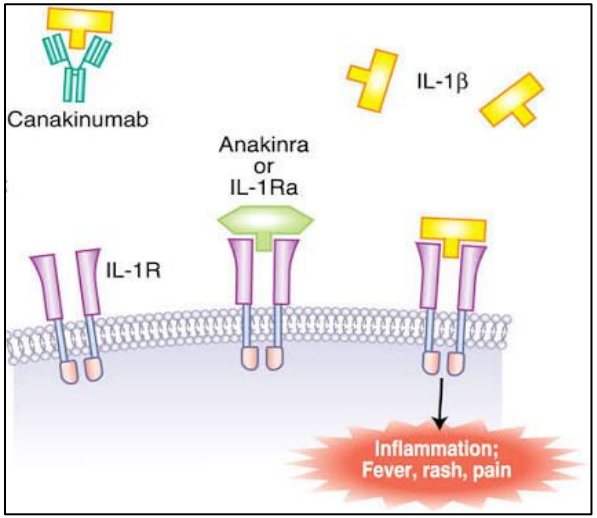
Read more here

HLA-B*58:01 allele: a genetic variant strongly linked to higher risk of life-threatening severe cutaneous adverse reactions (SCARs) when taking allopurinol:

- Stevens-Johnson syndrome (SJS) - *shown in image*
- Toxic epidermal necrolysis (TEN)
- Drug reaction with eosinophilia and systemic symptoms (DRESS)

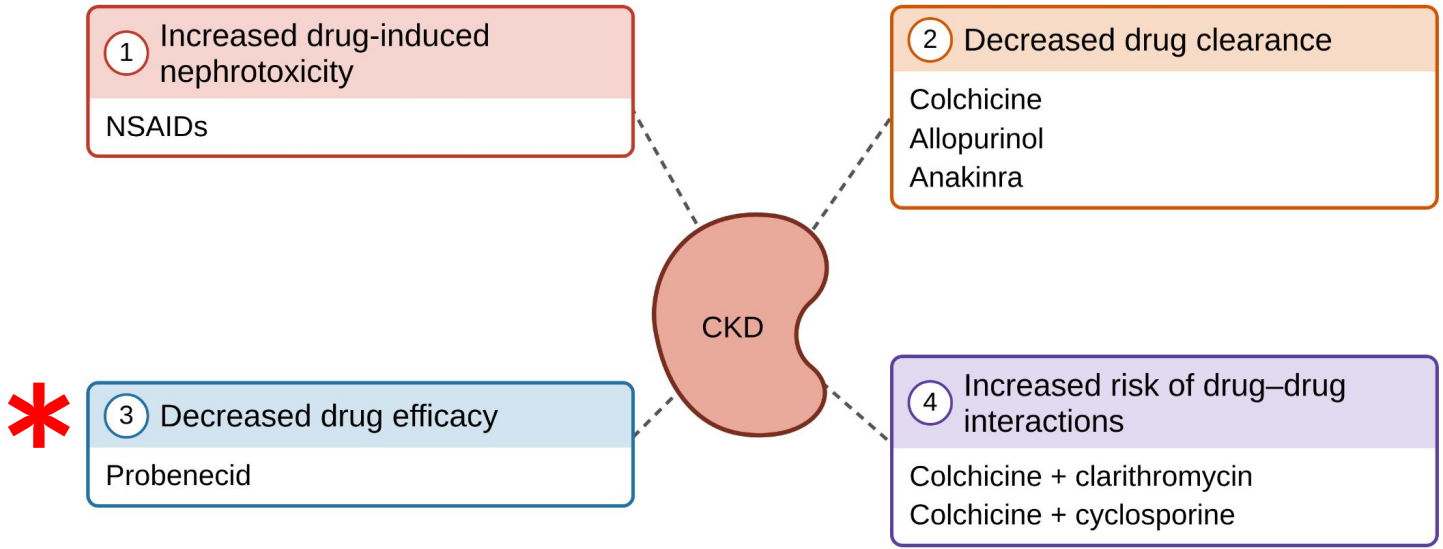
Clearance of IL-1 Inhibitors: Anakinra vs. Canakinumab

	Anakinra	Canakinumab
MOA	IL-1 receptor antagonist	Monoclonal antibody against IL-1β
Gout indication	Off-label	FDA-approved (2023) when NSAIDs/colchicine are unsuitable and repeated steroids are inappropriate
Kidney	Cleared by the kidney; accumulates in CKD	Largely unaffected
Key adverse effects	Injection-site reactions, infection, neutropenia	Infection; long-acting, so it can't be stopped quickly
Contraindications	Active infection; live vaccines	Active infection; live vaccines



Same pathway, different clearance.
In CKD, anakinra accumulates while canakinumab is largely unaffected.

How CKD Can Affect the Safety and Efficacy of Gout Drugs

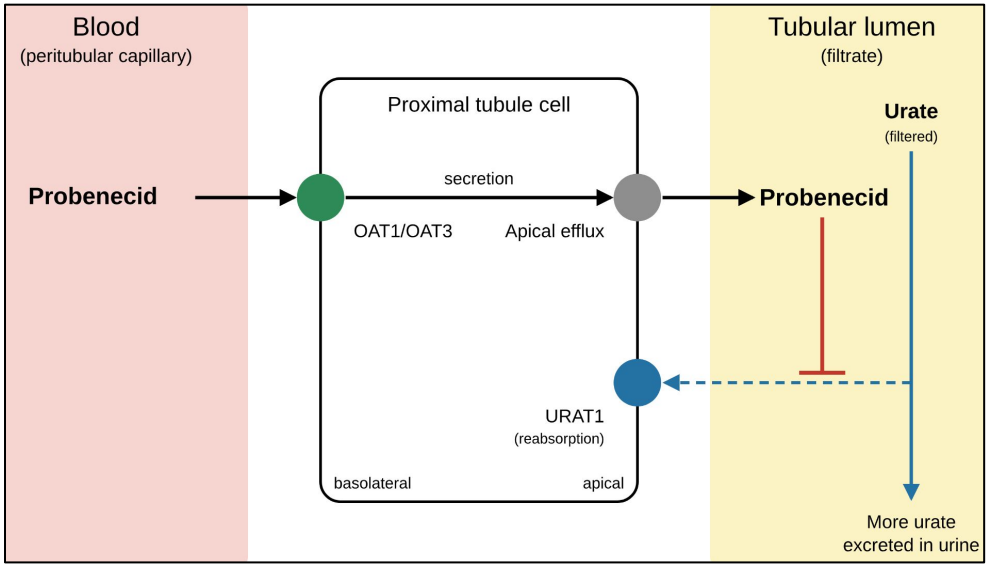


Drugs that are largely unaffected by reduced kidney function

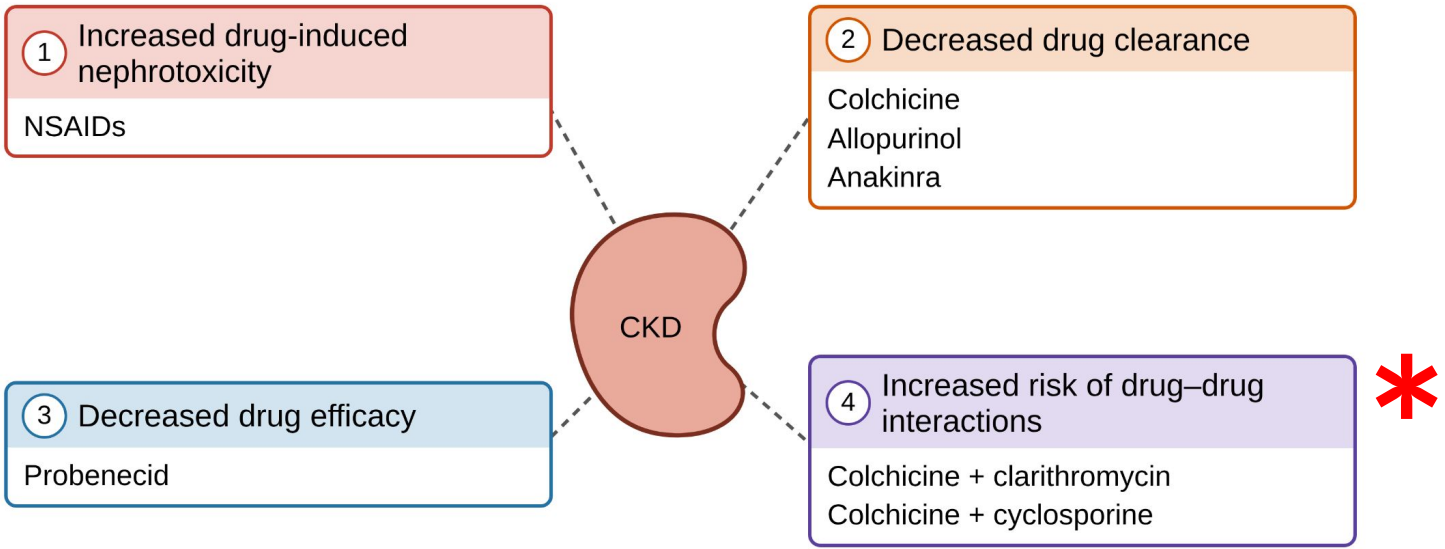
Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

Decreased Drug Efficacy: Probenecid

- Probenecid is a uricosuric urate-lowering drug.
- MOA (review): inhibits **URAT1** in the proximal tubule, thereby inhibiting urate reabsorption. It reaches the tubule by secretion from the blood.
- In CKD, reduced blood flow to the tubules means less probenecid is delivered to its site of action, and reduced GFR means less urate is filtered, so **efficacy decreases**.
- Ineffective in CKD stage 4–5; xanthine oxidase inhibitors are preferred in CKD.
- Adverse effect: **uric acid kidney stones** (avoid in patients with a history of uric acid stones).
- Aspirin reduces its effect.



How CKD Can Affect the Safety and Efficacy of Gout Drugs

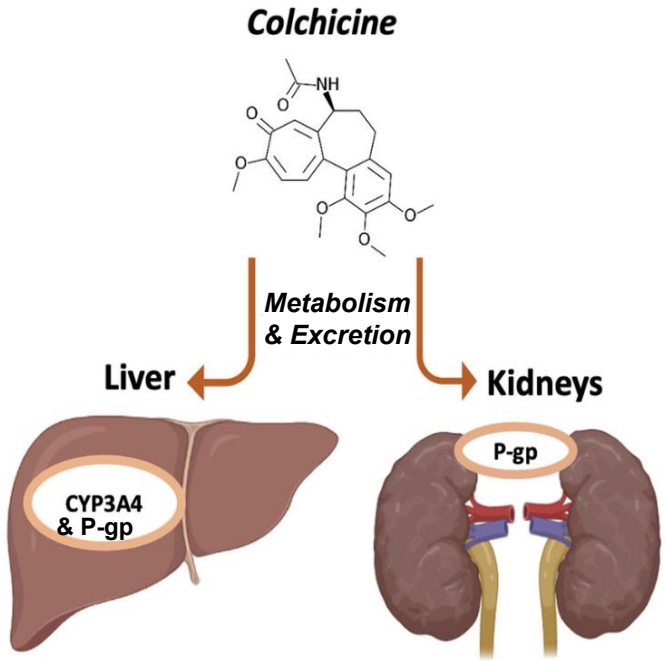


Drugs that are largely unaffected by reduced kidney function

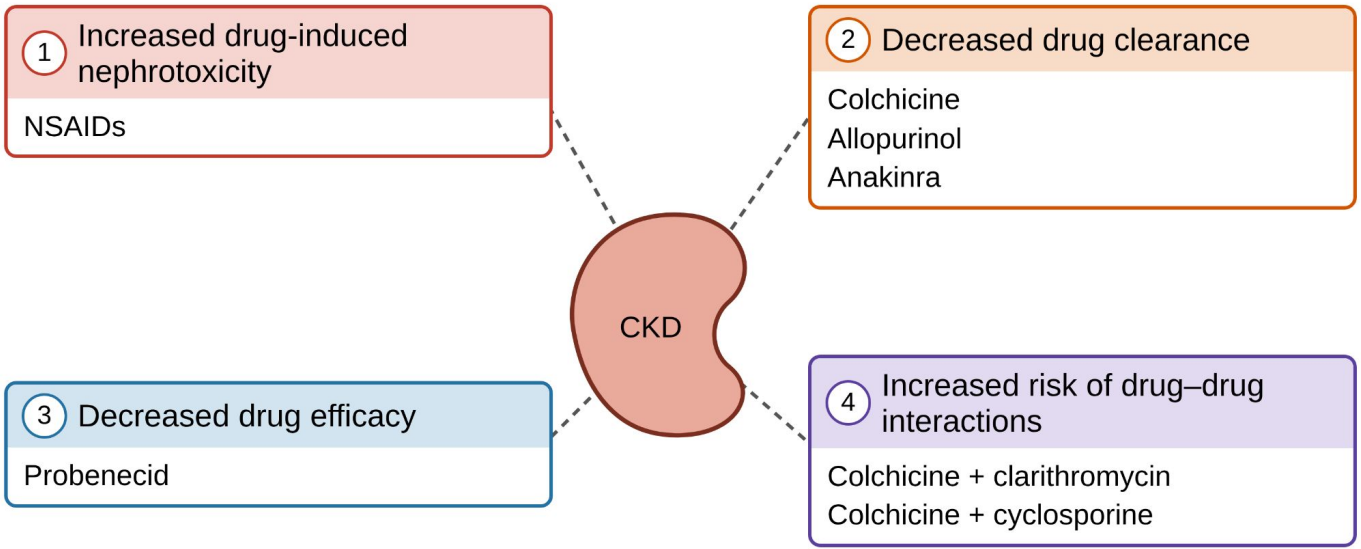
Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

Increased Risk of Drug-Drug Interactions: Colchicine

- Colchicine is eliminated by **hepatic metabolism (CYP3A4)** and by **P-gp efflux** into bile (in liver) and urine (in kidneys).
- In CKD, renal elimination is reduced.
- Clarithromycin and cyclosporine inhibit CYP3A4 and P-gp, so **colchicine levels rise and toxicity increases.**
- **Contraindicated:** colchicine + strong CYP3A4 or P-gp inhibitors in patients with renal or hepatic impairment.
- Statins and fibrates can also cause **myopathy**, adding to colchicine-induced muscle toxicity (myoneuropathy, rhabdomyolysis).



How CKD Can Affect the Safety and Efficacy of Gout Drugs

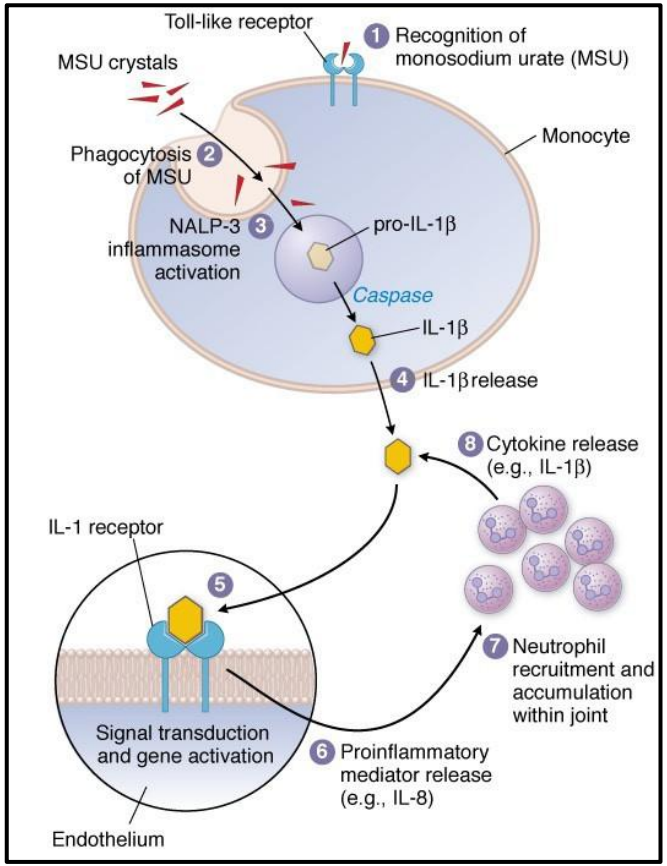


* Drugs that are largely unaffected by reduced kidney function

Glucocorticoids · Febuxostat · Canakinumab · Pegloticase

Drugs Largely Unaffected by CKD: Glucocorticoids

- MOA (review): reduce inflammatory gene expression
- Metabolized by the liver, so they're largely unaffected by CKD. **Often the preferred flare treatment in CKD.**
- Route: intra-articular for 1-2 joints; oral for multiple joints.
- Adverse effects: hyperglycemia, fluid retention, hypertension, infection.
    - With repeated or prolonged use: osteoporosis, impaired wound healing, and adrenal suppression.
- Contraindication: **septic arthritis** (exclude infection before injecting a joint).



Drugs Largely Unaffected by CKD: Febuxostat

Goal	Class	Drugs	Mechanism
Treat flares and prevent flares when starting urate lowering	**NSAIDs**	Naproxen, indomethacin, ibuprofen	Inhibit COX-1/2; ↓ prostaglandins
	Colchicine	Colchicine	Binds tubulin; ↓ neutrophil migration and ↓ NLRP3 inflammasome activation
	Glucocorticoids	Prednisone, prednisolone, triamcinolone (intra-articular)	↓ inflammatory gene expression
	IL-1 inhibitors	Anakinra (off-label), canakinumab	Block IL-1 signaling (anakinra: IL-1 receptor; canakinumab: IL-1β)
Lower urate (recurrent gout, tophi)	**Xanthine oxidase inhibitors**	Allopurinol, febuxostat *	Inhibit xanthine oxidase; ↓ uric acid production
	Uricosuric	Probenecid	Inhibits URAT1; ↑ urate excretion
	Uricase	Pegloticase	Converts uric acid to allantoin

Drugs Largely Unaffected by CKD: Febuxostat *(cont')*

	Allopurinol	Febuxostat
MOA	Purine analog; its active metabolite **oxypurinol** inhibits xanthine oxidase	Non-purine inhibitor of xanthine oxidase
Role in gout	**First-line** urate-lowering drug, including in CKD	**Alternative** when allopurinol is ineffective or not tolerated
Kidney	Oxypurinol is cleared by the kidney, so it **accumulates in CKD**. Start low, increase gradually.	**Metabolized by the liver**, so it's largely unaffected by CKD
Key adverse effects	Rash; **hypersensitivity syndrome** (risk increased by CKD and HLA-B*58:01)	Elevated liver enzymes; **boxed warning: increased cardiovascular death** compared with allopurinol
Contraindications / interactions	Prior hypersensitivity reaction; avoid with azathioprine/6-MP	**Contraindicated** with azathioprine/6-MP

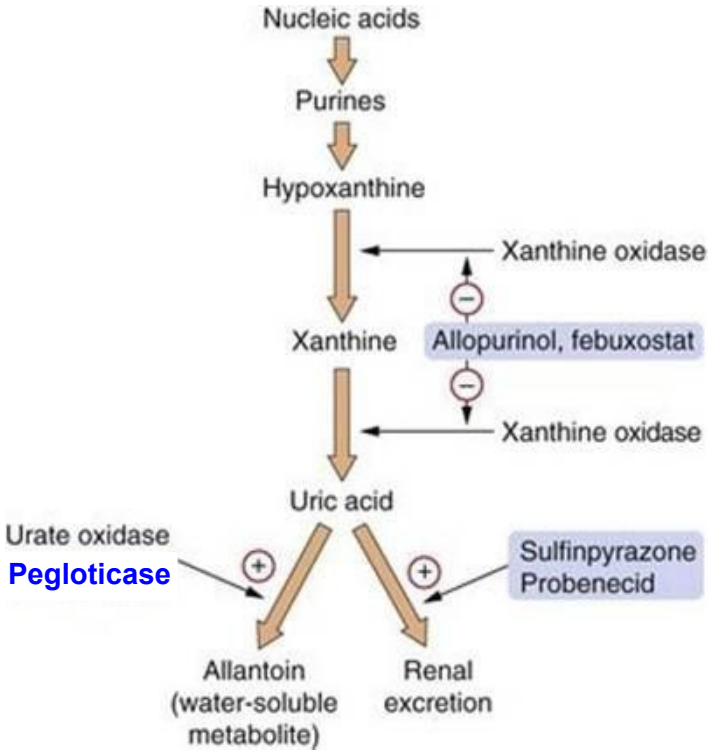
Same target, different clearance.
Allopurinol is still preferred in CKD, but febuxostat is an option when it can't be used.

Drugs Largely Unaffected by CKD: Pegloticase

Goal	Class	Drugs	Mechanism
Treat flares and prevent flares when starting urate lowering	**NSAIDs**	Naproxen, indomethacin, ibuprofen	Inhibit COX-1/2; ↓ prostaglandins
	Colchicine	Colchicine	Binds tubulin; ↓ neutrophil migration and ↓ NLRP3 inflammasome activation
	Glucocorticoids	Prednisone, prednisolone, triamcinolone (intra-articular)	↓ inflammatory gene expression
	IL-1 inhibitors	Anakinra (off-label), canakinumab	Block IL-1 signaling (anakinra: IL-1 receptor; canakinumab: IL-1β)
Lower urate (recurrent gout, tophi)	**Xanthine oxidase inhibitors**	Allopurinol, febuxostat	Inhibit xanthine oxidase; ↓ uric acid production
	Uricosuric	Probenecid	Inhibits URAT1; ↑ urate excretion
	Uricase	Pegloticase *	Converts uric acid to allantoin

Drugs Largely Unaffected by CKD: Pegloticase *(cont')*

- MOA (review): PEGylated recombinant uricase converts uric acid to allantoin, which is more soluble and easily excreted.
- Given as an IV infusion; works regardless of kidney function, so it's **largely unaffected by CKD.**
- Indication: severe gout **refractory to oral urate-lowering therapy.**
- Adverse effects: **anaphylaxis and infusion reactions**; gout flares when starting therapy.
  - Antibodies to the drug can cause loss of response, which increases the risk of infusion reactions.
- **Contraindication: G6PD deficiency.** The reaction produces hydrogen peroxide, which can cause hemolysis and methemoglobinemia.



Putting It Together: Choosing Flare Treatment in CKD

Drug	CKD stage 3	CKD stage 4–5 or dialysis
NSAIDs	Avoid	Avoid
Colchicine	Use with caution	Use with great caution; avoid repeated courses
Glucocorticoids	Preferred option	**Preferred option**
IL-1 inhibitors	Selected cases	Canakinumab is unaffected; anakinra accumulates

Putting It Together: Choosing Urate-Lowering Therapy in CKD

- **Indications:** ≥2 flares per year, tophi, or joint damage on imaging; *consider after a first flare in CKD stage 3 or worse.*
- **Goal:** serum urate **below 6 mg/dL.**
- **First-line:** allopurinol (start low, increase gradually, watch for hypersensitivity)
- **Alternative:** febuxostat
- **Avoid** probenecid in CKD stage 4–5 (ineffective); not preferred in stage 3.
- **Refractory gout:** pegloticase
- **Prevent early flares** with anti-inflammatory prophylaxis (not NSAIDs in CKD).
- **Don't treat** asymptomatic hyperuricemia to "protect the kidney"; trials showed no benefit.

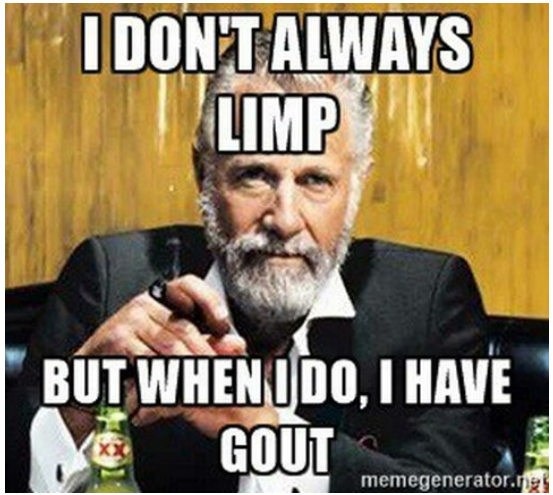
Mr. R is a 61-year-old man with CKD stage 4. Over the past year, his serum creatinine has been stable at about 2.5 mg/dL (0.6–1.2 mg/dL), and his serum urate has ranged from 9 to 10 mg/dL (3.0–8.2 mg/dL). He has had two gout flares in the past year, each treated with a short course of oral prednisone. Three days ago he developed a third flare and has been taking over-the-counter ibuprofen for the pain. On examination, his right first metatarsophalangeal joint is red, warm, and swollen, and a small tophus is present on his left elbow. Today his serum creatinine is 3.3 mg/dL, indicating worsening kidney function.

Why did his creatinine rise?

How would you treat this flare?

Should he start urate-lowering therapy?

Questions?


I DON'T ALWAYS
LIMP
BUT WHEN I DO, I HAVE
GOUT
memegenerator.net